

2014 F=ma Exam: Problem 4

Kevin S. Huang

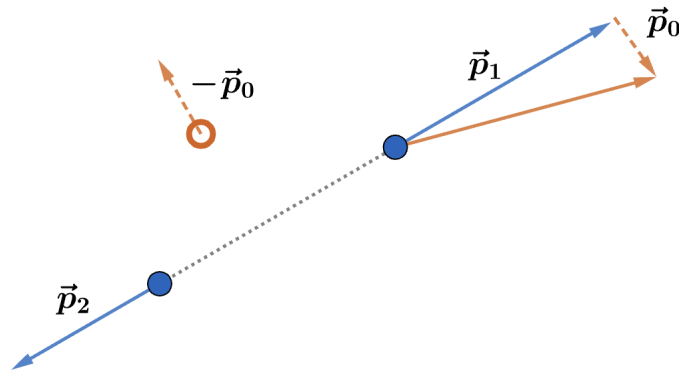
1st statement: If a particle at rest explodes into 2 particles, by conserving momentum and energy we obtain

$$\begin{aligned}\vec{p}_1 + \vec{p}_2 &= 0 \\ K_1 + K_2 &= \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} = K\end{aligned}$$

The first equation tells us the magnitudes of the momenta are equal $p_1 = p_2 \equiv p$ so

$$\frac{p^2}{2} \left(\frac{1}{m_1} + \frac{1}{m_2} \right) = K$$

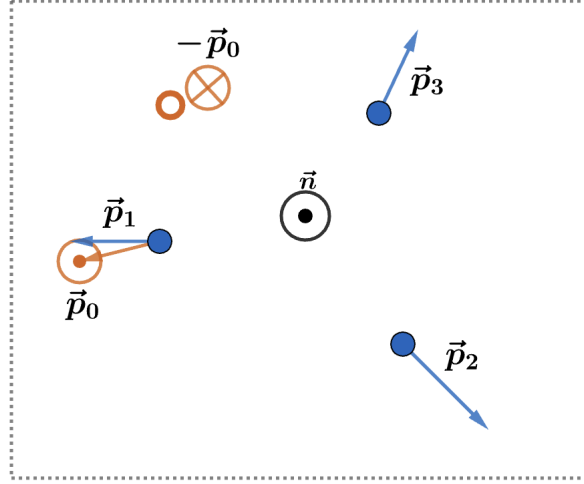
which can be solved for p and then we know K_1 and K_2 . In this case, we have two equations for two unknowns p_1, p_2 . For three particles, we still have these two equations for three unknowns p_1, p_2, p_3 so there is not a unique solution. Hence, $N_1 = 2$.



2nd statement: If a particle at rest explodes into 2 particles, then by conservation of momentum,

$$\begin{aligned}\vec{p}_1 + \vec{p}_2 &= 0 \\ \vec{p}_1 &= -\vec{p}_2 \\ \vec{v}_1 &\propto -\vec{v}_2\end{aligned}$$

so the velocities of the particles lie in a line (special case of lying in a plane). If we added some momentum $\vec{p}_0$ (perpendicular to the line) to the first particle, then we could restore momentum conservation by adding a third particle with momentum $-\vec{p}_0$. Thus, this counterexample illustrates the velocities of 3 particles do not have to lie in a line. Hence, $N_2 = 2$.



3rd statement: If a particle at rest explodes into 3 particles, then by conservation of momentum,

$$\vec{p}_1 + \vec{p}_2 + \vec{p}_3 = 0$$

$$\vec{p}_1 = -\vec{p}_2 - \vec{p}_3$$

The plane containing $\vec{p}_2$ and $\vec{p}_3$ can be defined by the perpendicular vector $\vec{n} = \vec{p}_2 \times \vec{p}_3$. Using the fact that $\vec{n} \cdot \vec{p}_2 = \vec{n} \cdot \vec{p}_3 = 0$, we have

$$\vec{n} \cdot \vec{p}_1 = \vec{n} \cdot -(\vec{p}_2 + \vec{p}_3) = -\vec{n} \cdot \vec{p}_2 - \vec{n} \cdot \vec{p}_3 = 0$$

so $\vec{p}_1$ is also perpendicular to $\vec{n}$ and thus in the plane of $\vec{p}_2$ and $\vec{p}_3$. Hence, the velocities of the particles lie in a plane. If we added some momentum $\vec{p}_0$ (perpendicular to the plane) to the first particle, then we could restore momentum conservation by adding a fourth particle with momentum $-\vec{p}_0$. Thus, this counterexample illustrates the velocities of 4 particles do not have to lie in a plane. Hence $N_3 = 3$.

Combining our results, the answer is C.